



Catholic Junior College
JC 2 Preliminary Examinations
Higher 1

CHEMISTRY

8873/01

Paper 1 Multiple Choice

16 September 2021

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, HT group and NRIC/FIN number on the Answer Sheet in the spaces provided.

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

WORKED SOLUTIONS

For each question there are **four** possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1 The relative abundances of the isotopes of a sample of krypton are shown in the table below.

Relative Isotopic Mass	82	83	84	86
Percentage Abundance	7.8	8.6	76.0	7.6

What is the relative atomic mass of krypton in this sample, given to 2 decimal places?

A 83.91

B 83.90

C 83.89

D 83.88

Concept: The Mole Concept and Stoichiometry. Calculate the relative atomic mass of an element given the relative abundances of its isotopes

Answer: A

Relative atomic mass of krypton

$$\begin{aligned}
 &= \frac{\sum (\text{relative isotopic mass} \times \text{relative abundance})}{\sum \text{relative abundance}} \\
 &= \frac{82 \times 7.8 + 83 \times 8.6 + 84 \times 76.0 + 86 \times 7.6}{7.8 + 8.6 + 76.0 + 7.6} \\
 &= 83.91 \text{ (2.d.p.)}
 \end{aligned}$$

- 2 Carbon disulfide vapour, $\text{CS}_2(\text{g})$, burns in oxygen to give $\text{CO}_2(\text{g})$ and $\text{SO}_2(\text{g})$. A sample of 20 cm^3 of $\text{CS}_2(\text{g})$ was completely burned in 100 cm^3 of oxygen. The volume of gas after burning was measured. This gaseous mixture was then treated with an excess of $\text{KOH}(\text{aq})$ and the volume of gas measured again. All measurements were made at the same temperature and pressure; under such conditions that carbon disulfide was gaseous.

What were the measured volumes?

	Volume of gas after burning/ cm^3	Volume of gas after adding $\text{KOH}(\text{aq})$ / cm^3
A	60	0
B	60	40
C	100	40
D	10	100

Concept: The Mole Concept and Stoichiometry. Perform calculations, including use of the mole concept, involving volumes of gases

Answer: C



	$\text{CS}_2(\text{g}) + 3\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{SO}_2(\text{g})$			
Initial / cm³	20	100	0	0
Change / cm³	-20	-60	+20	+40
Final / cm³	0	40	20	40

Total volume of gas = 0 + 40 + 20 + 40 = 100 cm³ of gas

Acidic gases are removed when the mixture of gases passes through KOH.

Hence CO₂ and SO₂ are removed. What is left is 40 cm³ of oxygen gas.

- 3** Which one of the following has the same number of particles as one mole of magnesium atoms?

- A** the number of hydrogen atoms in 18.0 g of water
- B** the number of ions in 2 dm³ of 0.25 mol dm⁻³ of aqueous nitric acid
- C** the number of delocalised electrons in two moles of copper metal
- D** the number of molecules in 35.5 g of chlorine gas

Concept: The Mole Concept and Stoichiometry. Define the term mole in terms of the Avogadro constant.

Answer: B

- A** 18.0 g of water is 1 mole of water. Each water molecule has 2 hydrogen atoms. Total number of moles of hydrogen atoms = 1 x 2 = **2 moles**.
- B** 2 dm³ of 0.25 mol dm⁻³ of aqueous nitric acid is 0.5 moles of nitric acid. $\text{HNO}_3 \rightarrow \text{H}^+ + \text{NO}_3^-$ Every mole of nitric acid fully dissociates into two moles of ions. Hence total number of moles of ions = 0.5 x 2 = **1 mole**
- C** $\text{Cu} \rightarrow \text{Cu}^{2+} + 2\text{e}^-$. Each mole of copper metal gives two moles of electrons. Hence two moles of copper metal give 2 x 2 = **4 moles of electrons**
- D** 35.5 g of chlorine gas ($M_r = 71.0$) = 0.5 moles of gas = **0.5 moles of gas molecules**.

4 Use of the Data Booklet is relevant to this question.

The ^{68}Ge isotope of the Group 14 element germanium is medically useful because it undergoes a natural radioactive process to give a gallium isotope, ^{68}Ga , which can be used to detect tumors. This transformation of germanium occurs when one of its electrons enters the nucleus, changing a proton into a neutron.

What do the isotopes ^{68}Ga and ^{68}Ge have in common?

- A Both isotopes have 37 neutrons in their nuclei.
- B Both isotopes have more electrons than protons.
- C Both isotopes have an outer electronic configuration $4s^2 4p^3$.
- D Both isotopes contain the same number of nucleons in their nuclei**

Concept: Atomic Structure. Deduce the numbers of protons, neutrons and electrons present in both atoms and ions given proton and nucleon numbers (and charge).

Answer: D

- A False, ^{68}Ge has 36 neutrons while ^{68}Ga has 37 neutrons.
- B False, both isotopes are electrically neutral (i.e. uncharged) hence they have equal numbers of electrons and protons.
- C False, ^{68}Ge has outer configuration of $4s^2 4p^2$ while ^{68}Ga has outer configuration $4s^2 4p^1$.
- D Correct. Nucleons = total no. of protons and neutrons = 68 for each element.**

5 Which of the following ions will be deflected by the largest angle in an electric field?

- A $^{23}\text{Na}^+$
- B $^{16}\text{O}^{2-}$**
- C $^{27}\text{Al}^{3+}$
- D $^{19}\text{F}^-$

Concept: Atomic Structure. Deflection of particles in an electric field.

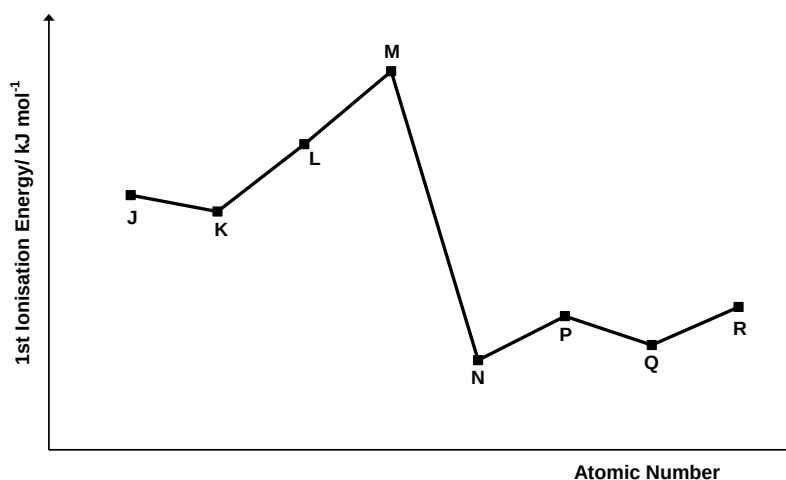
Answer: B

angle of deflection $\propto \frac{\text{charge}}{\text{mass}}$ of particle

A	$^{23}\text{Na}^+$	B	$^{16}\text{O}^{2-}$	C	$^{27}\text{Al}^{3+}$	D	$^{19}\text{F}^-$
$\frac{\text{ionic charge}}{\text{mass}} \propto 0.043$		0.125		0.111		0.053	

$^{16}\text{O}^{2-}$ has the largest ionic charge/mass ratio.

- 6 The following graph shows the first ionisation energies of eight consecutive elements **J** to **R**, which have atomic numbers between 3 to 20 in the Periodic Table.



Which one of the following statements about the elements is correct?

- A** Elements **N** and **K** can form the compound **N₂K**.
- B** Element **M** is a solid at room temperature.
- C** Aqueous solution of the chloride of element **P** is basic.
- D** Elements **J** and **R** belong to the same group.

Concept: The Periodic Table. First IE and identifying elements.

Answer: A

The large drop in first IE between **M** and **N** indicates that the removal of the valence electron from **M** is from a shell nearer the nucleus (period 2. Group 18, Ne) compared to the removal of a valence electron from **N** (Period 3. Group 1, Na).

- A** **Correct.** Elements **N** (Na) and **K** (O) can form the compound **N₂K** (**Na₂O**).
- B** **Incorrect.** Element **M** (Ne) is a gas at room temperature.
- C** **Incorrect.** Chloride of element **P** is acidic. **MgCl₂** undergoes slight hydrolysis.
- D** **Incorrect.** Element **J** is from Group 15 and **R** is from Group 14.

- 7 Use of the Data Booklet is relevant to this question.

Which of the following species has the most number of unpaired electrons?

A Al

B Si

C P^{3-}

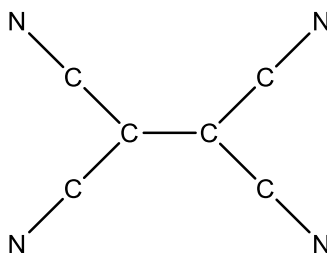
D Cr^{3+}

Concept: Atomic Structure. Electronic configuration.

Answer: D

Species	No. of unpaired valence electrons
Al: $[Ne] 3s^2 3p^1$	1
Si: $[Ne] 3s^2 3p^2$	2
P^{3-} : $[Ne] 3s^2 3p^6$	0
Cr^{3+} : $[Ar] 3d^3$	3

- 8 Tetracyanoethene is used in the preparation of semiconductors. The structure of tetracyanoethene is given below (double and triple bonds are not shown).



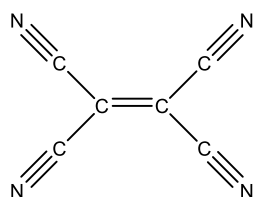
How many sigma, σ , and pi, π , bonds are there in the molecule?

	σ	π
A	9	0
B	9	1
C	9	5
D	9	9

Concept: Chemical Bonding. Sigma and pi bonding.

Answer: D

Full structural formulae:



Single bond contains 1 σ bond whereas triple bond contains 1 σ bond and 2 π bonds.

- 9 Which one of the following sequences shows the bonds arranged in order of increasing polarity?

- A H – O H – F H – Cl
 B H – O H – Cl H – F
 C H – Cl H – O H – F
 D H – Cl H – F H – O

Concept: Chemical Bonding. Bond Polarity.

Answer: C

The greater the difference in electronegativity between the two atoms, the more polar the bond. F is the most electronegative, followed by O and Cl. Hence the greatest electronegativity difference is between H and F resulting in the H-F bond being the most polar, followed by H-O and lastly, H-Cl.

- 10 Consider the following four molecules.

- 1 HF
 2 I₂
 3 H₂O
 4 CH₄

What is the order of **increasing** boiling point of the molecules?

- A 4 → 1 → 3 → 2 C 4 → 1 → 2 → 3
 B 4 → 2 → 1 → 3 D 1 → 4 → 3 → 2

Concept: Chemical Bonding. IMF.

Answer: A

Among the four compounds, iodine ($M_r = 254$) is the only solid at rtp hence it has the highest boiling point.

As for the remaining three compounds, HF and H₂O have higher boiling points than CH₄ since intermolecular hydrogen bonding in HF and H₂O are stronger than the intermolecular instantaneous dipole – induced dipole forces of attraction between CH₄ molecules.

11 Which molecules contain within their structures three atoms arranged in a straight line?

1 SO_2

2 HCN

3 XeF_4

A 2 only

B 1 and 2 only

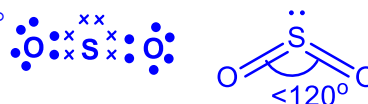
C 1, 2 and 3

D 2 and 3 only

Concept: Chemical Bonding. Shapes of molecules.

Answer: D

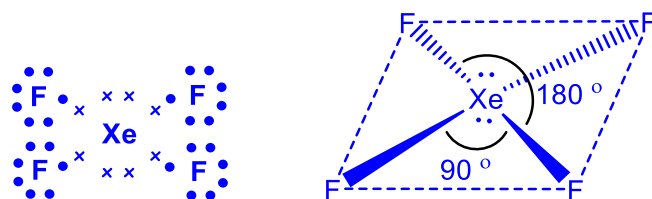
SO_2 : 2 bp and 1 lp $\Rightarrow$ bent, bond angle $< 120^\circ$



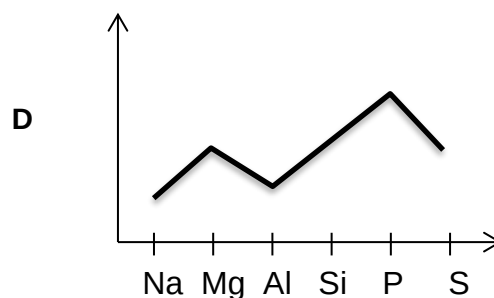
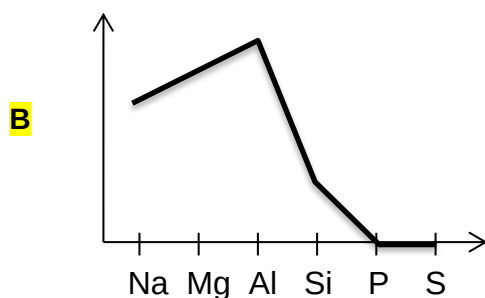
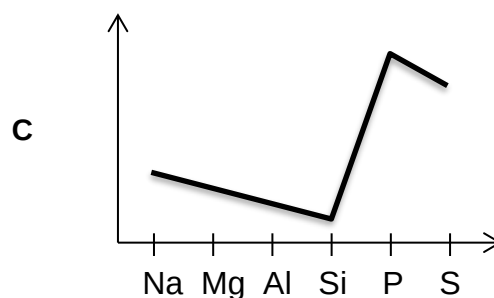
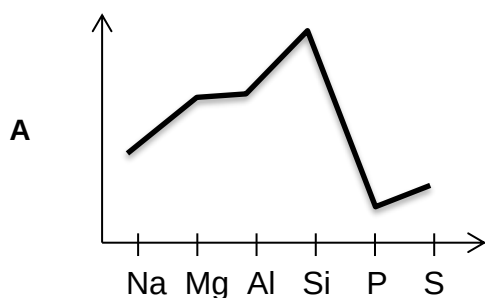
HCN : 2 bp and 0 lp $\Rightarrow$ linear, bond angle 180°



XeF_4 : 4 bp and 2 lp $\Rightarrow$ square planar, bond angle 90° and 180°



12 Which of the following graph shows how electrical conductivity of Period 3 elements varies?



Concept: The Periodic Table. Trends in physical properties.

Answer: B

Graph A shows the trend of melting point of Period 3 elements.

Graph B shows the trend of electrical conductivity of Period 3 elements.

Graph C shows the trend of ionic radii of Period 3 elements.

Graph D shows the trend of 1st IE of Period 3 elements.

- 13** The element astatine lies below iodine in Group 17 of the Periodic Table. Which one of the following properties is correct for astatine?

- A** It forms diatomic molecules which dissociate less readily than chlorine molecules.
- B** The hydride of astatine, HAt, is stable to heat.
- C** It can oxidise iodide to iodine.
- D** It exists as a crystalline solid at room temperature.

Concept: The Periodic Table. Group 17.

Answer: D

- A** **Incorrect.** It forms diatomic molecules which dissociate **more** readily than chlorine molecules. At-At bonds are longer and weaker than Cl-Cl bonds.
- B** **Incorrect.** H-F and H-Cl have higher bond energies and are stable to heat. HBr decomposes slightly and HI decomposes easily. Hence HAt is expected to decompose even more readily than HI.
- C** **Incorrect.** The oxidising power of the halogens can be illustrated by displacement reactions. A more reactive halogen will displace a less reactive one from its compounds. At is less reactive and hence cannot oxidise iodide to iodine.
- D** **Correct.** Volatility of Group 17 elements, X₂ decreases down the group. Down the group, number of electrons in the molecule increases. Hence, the strength of instantaneous dipole-induced dipole attraction between X₂ molecules increases. Energy needed to overcome the stronger intermolecular forces increases. Since I₂ is already a solid at rtp, At₂ is expected to be a solid too.

- 14 Lithium has one valence electron and belongs to Group 1 of the Periodic Table. Which of the following statements are true about lithium?

- 1 Lithium has the smallest radius in Group 1.
- 2 Lithium is the weakest reducing agent in Group 1.
- 3 Lithium is the most chemically reactive in Group 1.

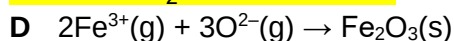
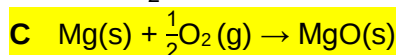
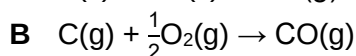
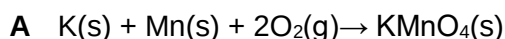
A 1 only B 3 only **C 1 and 2 only** D 1, 2 and 3 only

Concept: The Periodic Table. Group 1.

Answer: C

Lithium's tendency to lose the valence electron is the lowest, and therefore the reducing power (tendency to be oxidised) and reactivity are the lowest among the Group 1 elements.

- 15 Which of the following is an equation that correctly defines a standard enthalpy change of combustion?



Concept: Chemical Energetics: Thermochemistry. Enthalpy change of combustion

Answer: C

ΔH_c° is the enthalpy when 1 mole of substance is burnt completely in oxygen under standard conditions of 298 K and 1 bar.

Equation A represents the ΔH_f° of KMnO_4 . There is more than 1 mole of substance undergoing reaction, thus it cannot be ΔH_c° .

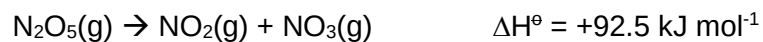
Equation B: Carbon is a solid at standard conditions. Also, the equation is represents incomplete combustion of carbon, thus it is not ΔH_c° .

Equation D represents lattice energy of Fe_2O_3 .

- 16 The following table lists some $\Delta H_f^\ominus$ values.

Compound	$\Delta H_f^\ominus / \text{kJ mol}^{-1}$
$\text{NO}_2(\text{g})$	+33.2
$\text{N}_2\text{O}_5(\text{g})$	+5.0

The enthalpy change for the dissociation of $\text{N}_2\text{O}_5(\text{g})$ is given below.

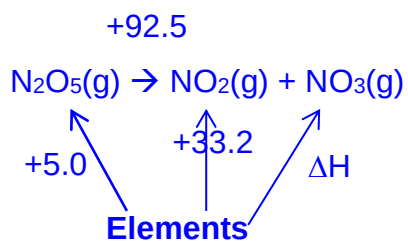


What is the standard enthalpy change of formation of $\text{NO}_3(\text{g})$?

- A - 130.7 kJ mol⁻¹ **C + 64.3 kJ mol⁻¹**
 B - 64.3 kJ mol⁻¹ D + 130.7 kJ mol⁻¹

Concept: Chemical Energetics: Hess' Law

Answer: C



Using Hess' Law, $+5.0 + 92.5 = +33.2 + \Delta H_f(\text{NO}_3)$

$\Delta H_f(\text{NO}_3) = + 64.3 \text{ kJ mol}^{-1}$

- 17 I₂ vapour is a purple gas while H₂ and HI are colourless gases.

An equilibrium is established between I₂, H₂ and HI in a closed vessel.

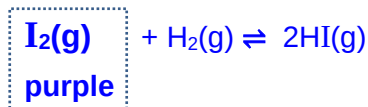


Which row describes the effects of changing conditions on the colour of the equilibrium mixture above?

	increasing the pressure	decreasing the temperature
A	no change in the colour intensity	colour becomes lighter
B	colour becomes darker	colour becomes lighter
C	colour becomes lighter	colour becomes darker
D	no change in the colour intensity	colour becomes darker

Concept: Chemical Equilibria; factors affecting position of equilibrium

Answer: A



Increasing the pressure has no effect on the position of equilibrium as the number of moles of gases on both sides of the equation is the same. Hence the colour intensity of the equilibrium remains unchanged.

Decreasing the temperature will favour the forward exothermic reaction hence the position of equilibrium shifts to the right, producing more colourless HI gas, resulting in the colour of the equilibrium mixture becoming lighter.

- 18 The following equilibrium exists in a sample of aluminium chloride vapour.



Which of the following will increase when the pressure is increased at constant temperature?

- A activation energy
- B amount of Al_2Cl_6 formed**
- C enthalpy change of reaction
- D rate constant for the forward reaction

Concept: Chemical Equilibria; factors affecting position of equilibrium

Answer: B

Option B: When pressure increases, the position of the equilibrium shifts to the left as the reverse reaction produces fewer number of moles of gaseous molecules. Hence the amount of Al_2Cl_6 formed increases.

Options A and D are incorrect: Activation energy and rate constant are independent of pressure. They temperature dependent (will increase as temperature increases), hence they will remain unchanged as temperature remains constant.

Option C: Enthalpy change is dependent on the amount of energy absorbed during bond breaking and energy evolved during bond formation and is independent of pressure.

- 19 Water is considered 'hard' when it is high in dissolved minerals such as calcium and magnesium. Hardness in tap water can be determined by titrating a sample of the water against a reagent which reacts with dissolved metal ions. The indicator for this titration requires the pH to be maintained at about 10.

Which substances, in aqueous solution, could be used to maintain the pH at about 10?

- A ammonia only
- B ammonia and sodium hydroxide
- C ammonia and ammonium chloride**
- D sodium hydroxide and sodium ethanoate

Concept: Theories of Acids & Bases, alkaline buffer

Answer: C

An alkaline buffer which consist of a weak base (ammonia) and a salt of the weak base (ammonium chloride) is required to maintain pH at an alkaline range of about 10.

- 20 The pH of human blood is constant at about 7.40.
Which ion or molecule present in the human body is responsible for removing $\text{H}^+(\text{aq})$ ions from the blood to keep the pH constant?

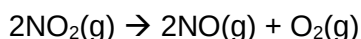
A NH_3 **B HCO_3^-** C CO_3^{2-} D PO_4^{3-}

Concept: Theories of Acids & Bases, $\text{H}_2\text{CO}_3/\text{HCO}_3^-$ buffer

Answer: B

$\text{H}_2\text{CO}_3/\text{HCO}_3^-$ buffer regulates the pH level in blood. HCO_3^- reacts with excess H^+ to form H_2O and CO_2 , hence keeping pH of blood relatively constant.

- 21 The decomposition of NO_2 is a second order reaction.



Which one of the following graphs is a straight line?

- A $[\text{NO}_2]$ against $(\text{rate of reaction})^2$
B $[\text{NO}_2]^2$ against (rate of reaction)
 C $[\text{NO}_2]$ against $(\text{time})^2$
 D $[\text{NO}_2]^2$ against (time)

Concept: Reaction Kinetics, graphs

Answer: B

$$\underbrace{\text{Rate}}_y = k \underbrace{[\text{NO}_2]^2}_x$$

To get a straight line graph, $y = mx$, then y-axis should be rate, and x-axis should be $[\text{NO}_2]^2$. However, if the axes are reversed, a straight line graph is also obtained.

- 22 An enzyme, **alcohol dehydrogenase**, can be used to oxidise ethanol but the enzyme is never used to oxidise butan-1-ol. Why is this so?

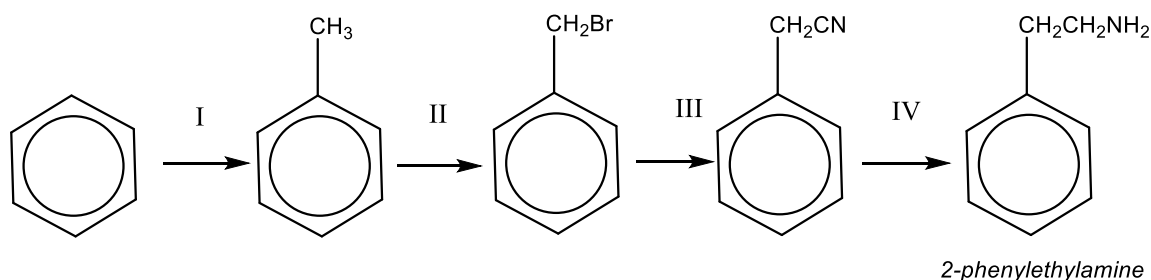
- A Ethanol can be oxidised easily whereas butan-1-ol is resistant to oxidation.
 B The enzyme is specific to the oxidation of ethanol and is not able to oxidise butan-1-ol.
 C There was sufficient enzyme to oxidise ethanol, but more enzyme is required before the oxidation of butan-1-ol occurs.
 D Ethanol is easily oxidised at room temperature whereas butan-1-ol requires heat before oxidation occurs.

Concept: Reaction Kinetics, Enzymes as specific catalysts

Answer: B

- A Incorrect. Ethanol and butan-1-ol can be oxidised easily.
 B Correct. Enzymes as biological catalysts which may have specific activity.
 C and D Have no information in the question that support the statements.

- 23 2-phenylethylamine is the 'feel-good' factor in chocolate. It can be synthesised from benzene in the following sequence.



Which set of information is true of the steps in the sequence?

	I	II	III	IV
	Type of reaction	Reagents and conditions	Type of reaction	Type of reaction
A	substitution	Br ₂ , uv light	substitution	reduction
B	substitution	HBr, heat	addition	addition
C	addition	Br ₂ , uv light	substitution	reduction
D	addition	HBr, heat	addition	addition

Concept: Polymers. Type of reactions, reaction of alkanes**Answer: A**

	I Type of reaction	II Reagents and conditions	III Type of reaction	IV Type of reaction
A	Substitution. A hydrogen atom on the benzene is replaced by the –CH ₃ group	Br ₂ , uv light. Free radical substitution of alkanes	Substitution. Br is replaced by a –C≡N	Reduction or addition of H ₂ are both correct. Hence all options for step IV are correct.

24 Which one of the following structural formulae exists in both *cis* and *trans* forms?

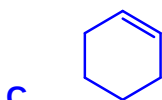
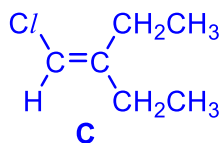
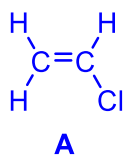
- A** CH₂=CHCl
B cyclohexene
C CHC≡C(CH₂CH₃)₂
D CH₃CH=CHCl

Concept: Polymers. Cis-trans isomers**Answer: D**

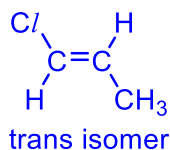
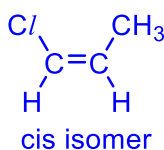
Criteria to form geometric isomers:

- Restricted rotation about C=C
- 2 different substituents attached to each carbon in the C=C

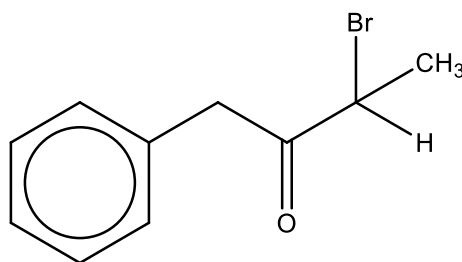
Options **A** and **C** are incorrect. There are 2 identical substituents on one of the carbon atoms in the C=C.



Cyclohexene only exists in the *cis* form. The *trans* form does not exist because the ring structure forces the two H atoms to be on the same side of the C=C double bond.

D

25 What are the functional groups present in the molecule shown below?



- A phenyl, aldehyde and tertiary alkyl bromide
- B phenyl, aldehyde and secondary alkyl bromide
- C phenyl, ketone and tertiary alkyl bromide
- D phenyl, ketone and secondary alkyl bromide**

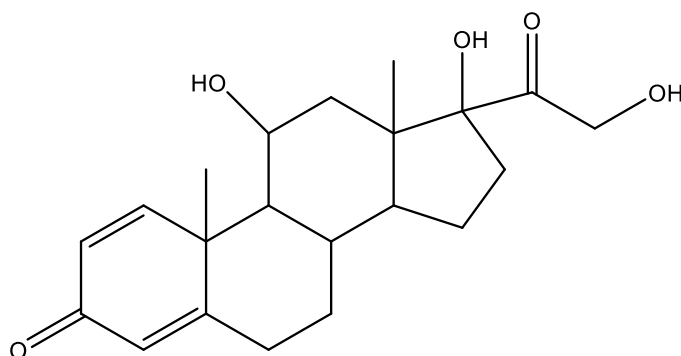
Concept: Polymers. Functional Groups

Answer: D

Knowledge of the phenyl group is irrelevant in this example.

The ketone group is obvious, as the aldehyde group requires a -CHO configuration. There may be a tendency to ignore the H and identify the functional group as a tertiary alkyl bromide, instead of secondary (with two alkyl groups only).

26 *Prednisolone* is a steroid medication used to treat certain types of allergies.



Prednisolone

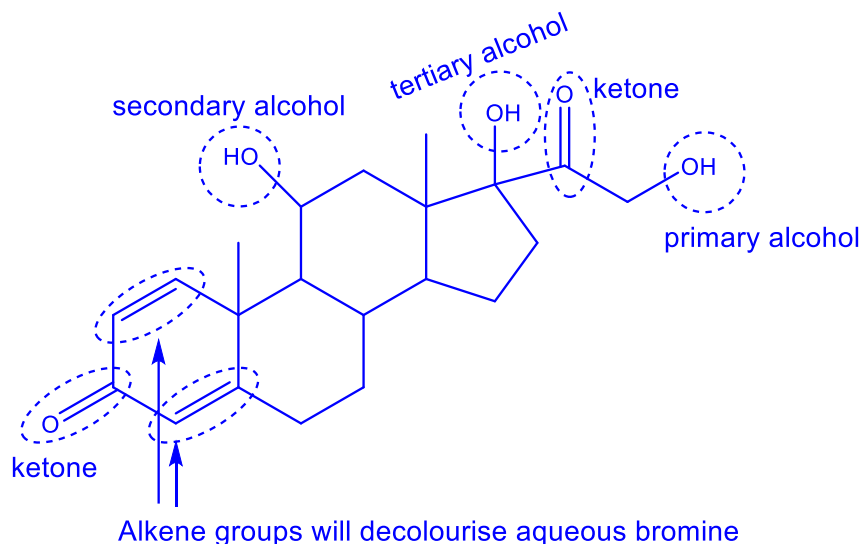
Which of the following is true of *Prednisolone*?

- 1 It has a carboxylic acid functional group.
- 2 It decolourises aqueous bromine.
- 3 It can undergo esterification upon reaction with $\text{CH}_3\text{CO}_2\text{H}$.
- 4 It will not react with NaBH_4 in methanol.

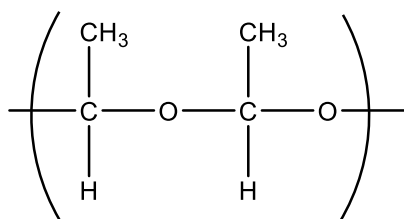
- A 2 only
- B 2 and 4 only
- C 1 and 3 only
- D 2 and 3 only**

Concept: Polymers, reaction of functional groups**Answer: D**

- 1 False. Refer to diagram below.
- 2 True. Alkene groups decolourise aqueous bromine.
- 3 True. Any of the alcohol groups can undergo esterification upon reaction with $\text{CH}_3\text{CO}_2\text{H}$.
- 4 False. There are two ketone groups that will undergo reduction with NaBH_4 .



- 27 Ethanal, $M_r = 44$, can be polymerised in the presence of a suitable reagent. The polymer has the following structure:



Which statements are correct?

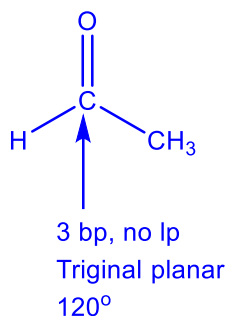
- 1 The relative molecular mass of the polymer molecule is an exact multiple of 44.
- 2 The polymer formed is a polyester.
- 3 The C-C-O bond angle changes from 120° in the monomer to 109° in the polymer.

A 1 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

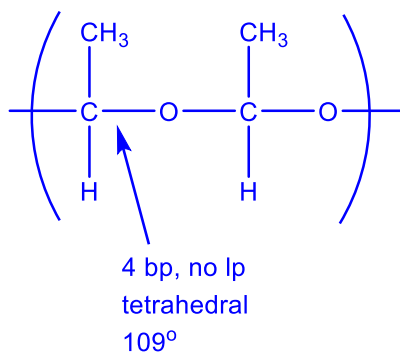
Concept: Polymers.**Answer: B**

1. True. This is an addition polymer. The monomers just add together giving a polymer whose M_r will be an exact multiple of the M_r of the monomer.
2. False. There is no ester linkage in the repeat unit.
3. True.

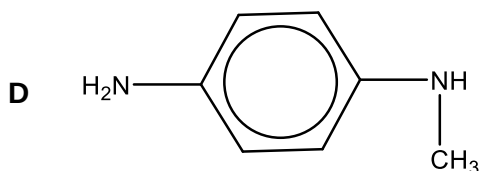
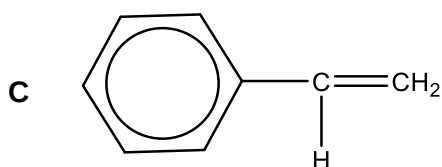
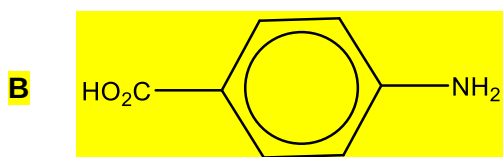
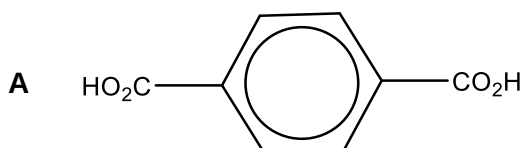
monomer:



repeat unit:



28 Which compound can form a condensation polymer by reacting with itself?

**Concept: Condensation polymers.****Answer: B**

Condensation polymers can be polyesters formed from carboxylic acid and alcohol group, or polyamides formed from carboxylic acid and amine groups (Option B)

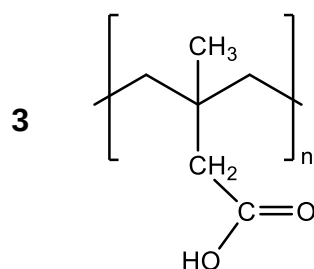
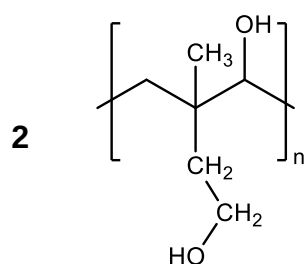
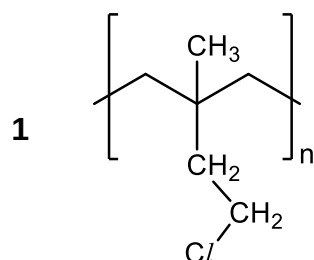
Option A has two carboxylic acid groups. A carboxylic acid group will not react with itself to form an ester.

Option C has an alkene group for addition polymerization.

Option D has two amine groups. These will not react to form a polymer.

- 29** Superabsorbent polymers (SAPs) are used to create hydrogels that are commonly used in baby diapers. Attempts were even made to use these hydrogels to plug the highly radioactive water leak after the Fukushima nuclear accident. These polymers have the ability to absorb large amounts of water and hold onto the water molecules for an extended period of time.

Which of the following is a likely polymeric structure of such SAPs?



- A** 1 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

Concept: Polymers. Properties.

Answer: C

To have the ability to absorb large amount of water, the polymer must be able to form hydrogen bonding. Only option 2 and 3 can form hydrogen bonding due to

the presence of a highly polarised H-X bond and at least one lone pair of electron on an electronegative atom X (where X = F, O or N). Option 1, although polar in nature, can only form permanent dipole – permanent dipole attraction.

- 30 Polymers are used extensively in our daily lives. Various polymers have found their way into products such as:

- soft contact lenses
- wrinkle free shirts
- microwavable food packaging
- bottles for alkaline cleaning solutions

Which row best describes the most suitable polymer to use for each type of product?

	Soft contact lenses	wrinkle free shirts	microwavable food packaging	bottles for alkaline cleaning solutions
A	poly(vinyl alcohol)	polyamide	LDPE	PET
B	poly(vinyl chloride)	polyester	poly(propene)	HDPE
C	poly(vinyl chloride)	polyamide	LDPE	PET
D	poly(vinyl alcohol)	polyester	poly(propene)	HDPE

Concept: Polymers. Uses of polymers

Answer: D

	Soft contact lenses	wrinkle free shirts	microwavable food packaging	bottles for alkaline cleaning solutions
D	poly(vinyl alcohol) The polymer needs to form hydrogen bonding with the moisture in the eyes so the eyes will not dry out.	polyester They form less hydrogen bonds than polyamides and are less prone to creasing.	poly(propene) Inert and no functional groups that will react with food.	HDPE Inert and no functional groups that will react with alkali. PET is not suitable as it is a polyester and it will hydrolyse.